Math Intro

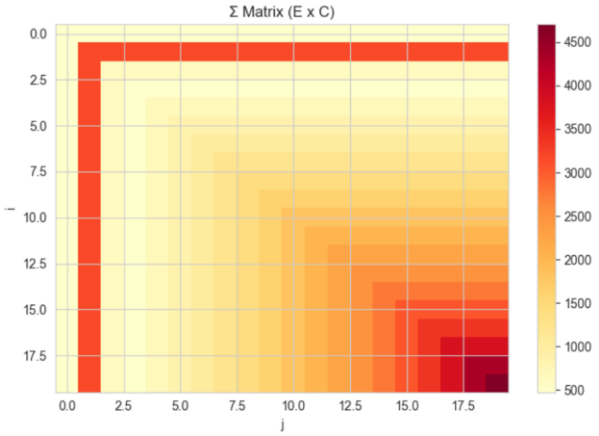
- **Filtration**: $U_1 \subset U_2 \subset \cdots \subset U_m$
- **Subadditive function**: $f : U \to \mathbb{R}_+$ with $f(U_1 \cup U_2) \leq f(U_1) + f(U_2)$
- **Inner product**: $\langle U_i, U_j \rangle = f(U_i) + f(U_j) - f(U_i \cap U_j)$
- **Matrix**: $\Sigma = (\langle U_i, U_j \rangle)_{i,j} = (f(U_{\min\{i,j\}}))_{i,j}$

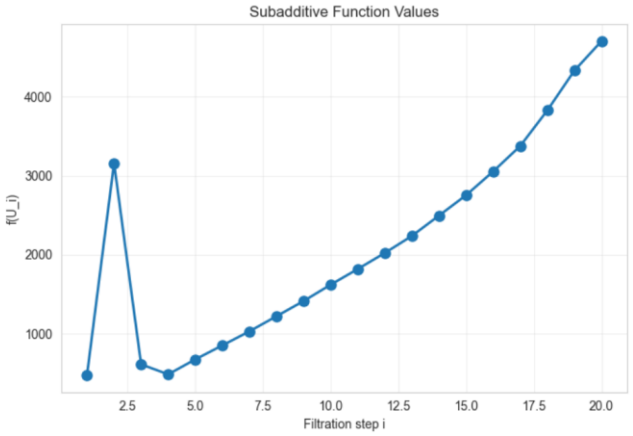
We take for **filtration** graphs $U_i = G_i$ built on one interval ordered by an increasing sequence of cutoffs.

For subadditive function $f : U \to \mathbb{Z}_+$ we take |Edges| x |Components| product.

Example

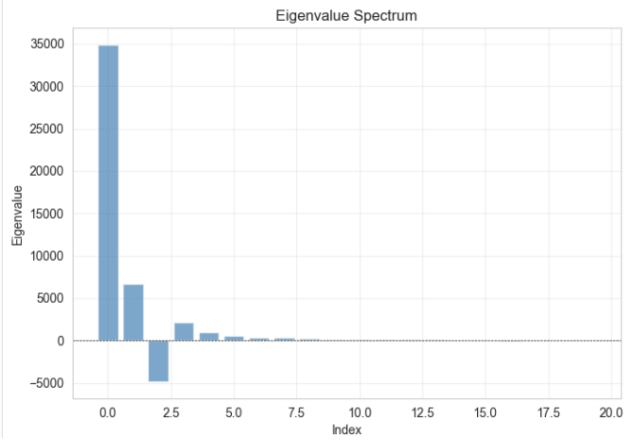
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f(U_i) values: [ 475. 3159.  614.  488.  677.  854. 1032. 1223. 1415. 1622. 1820. 2023. 2239. 2496. 2754. 3
054. 3376. 3826. 4330. 4702.]
```

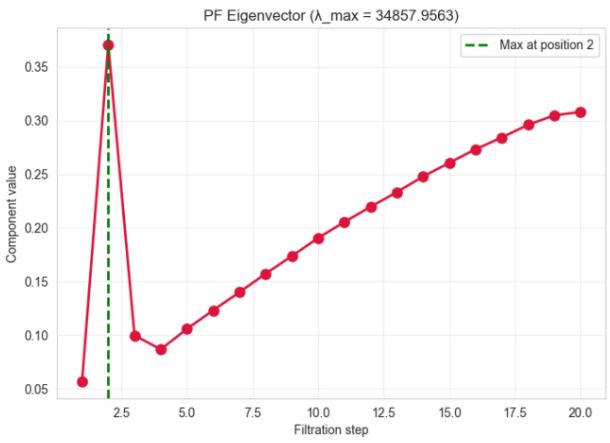




Perron-Frobenius Analysis (|Edge|*|Components|):

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Maximum eigenvalue λ_max = 34857.9563
Corresponding eigenvector with maximum component at position 2:
v_max = [0.05636483 0.37051578 0.09968577 0.08663762 0.1057401  0.12309279 0.13991496 0.15719907 0.1737078
0.19047474 0.20543074 0.21956807  0.23325018 0.24780965 0.26059161 0.27321162 0.28423331 0.29596696  0.3048
2936 0.30811756]
```



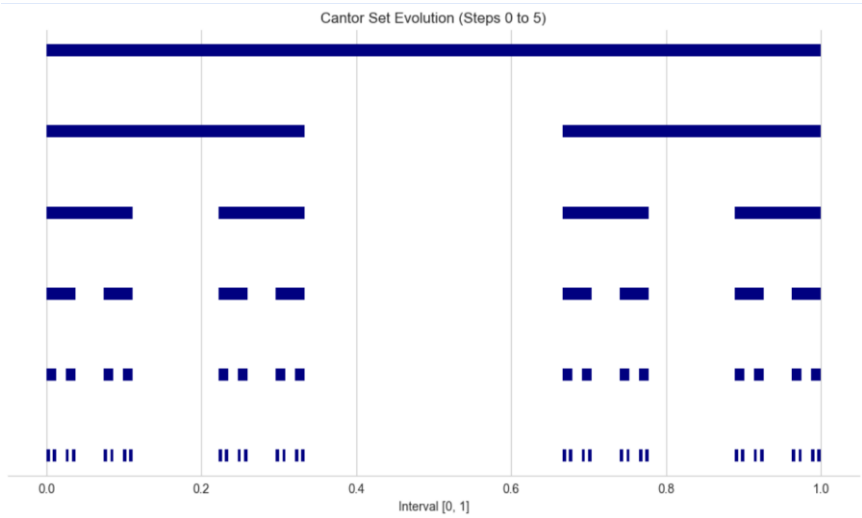


The PF eigenvector assigns a **weight to each scale (cutoff)**.

- Large vector components indicate scales that dominate
- A sharp peak suggests an *optimal cutoff*

The index of the max component of the PF eigenvector highlights, which cover resolution, contribute most.

Building Cantor Graphs



To produce a Cantor graph, the process begins by generating points on a middle-third **Cantor set** according to:

- **n_iterations:** controls the depth of the fractal construction.
- **Point Generation Strategy**
  - **Endpoints Only:** Generates exactly $2 \cdot 2^2$ points by using only the boundaries of each interval. This is the "cleanest" approach for conducting **connectivity tests**.
  - **Interior Sampling:** Generates $points_per_interval \cdot 2^2$ points (we sample evenly, but can be done randomly as well). By filling the intervals with interior points, we suppose to achieve better coverage for **fractal dimension estimation**.

Verifying Self-Similarity

To ensure the graph accurately reflects Cantor's recursive nature, self-similarity is verified across two distance regimes:

- **Endpoints Regime:** Using a distance cutoff of $\frac{1}{3^k} + \varepsilon$ recovers the structure of the previous (k-1) iteration.
- **Interior Sampling Regime:** The cutoff is min inner gap between points within an interval and the graph recovers the structure of the k-iteration of the Cantor set approximation.

Fractal Dimensions for Graphs

Hausdorff Dimension is a theoretical measure. It requires covering the set with balls of varying sizes and taking an infimum over all possible covers. This is generally computationally intractable to calculate directly for arbitrary data (finding the optimal ball covering is an NP-hard problem). Therefore, we use Minkowski–Bouligand box covering (grids) numerical approximation because it is computationally efficient and converges to the same value for self-similar fractals.

Minkowski–Bouligand (Box-Counting) Dimension D_{box} is defined by covering the set with a grid of boxes of size ϵ and counting how many boxes $N(\epsilon)$ contain at least one point of the set. Based on box-counting: $N(\epsilon) \sim \epsilon^{-D_{box}}$ where $N(\epsilon)$ is number of boxes of size ϵ needed to cover the set.

$$D_{box} = \lim_{\epsilon \to 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)}$$

For graphs, the number of balls (boxes) could be substituted with the number of edges $|E|$ and the fractal dimension approximation could be:

$$d_f \approx \frac{\log(|E|)}{\log(1/cutoff)}$$

Interpretation:

- Higher dimension → denser packing at given scale
- Captures scaling behavior of connectivity

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======================================================================
CANTOR SCALING ANALYSIS  (n=5,  endpoints only)
======================================================================

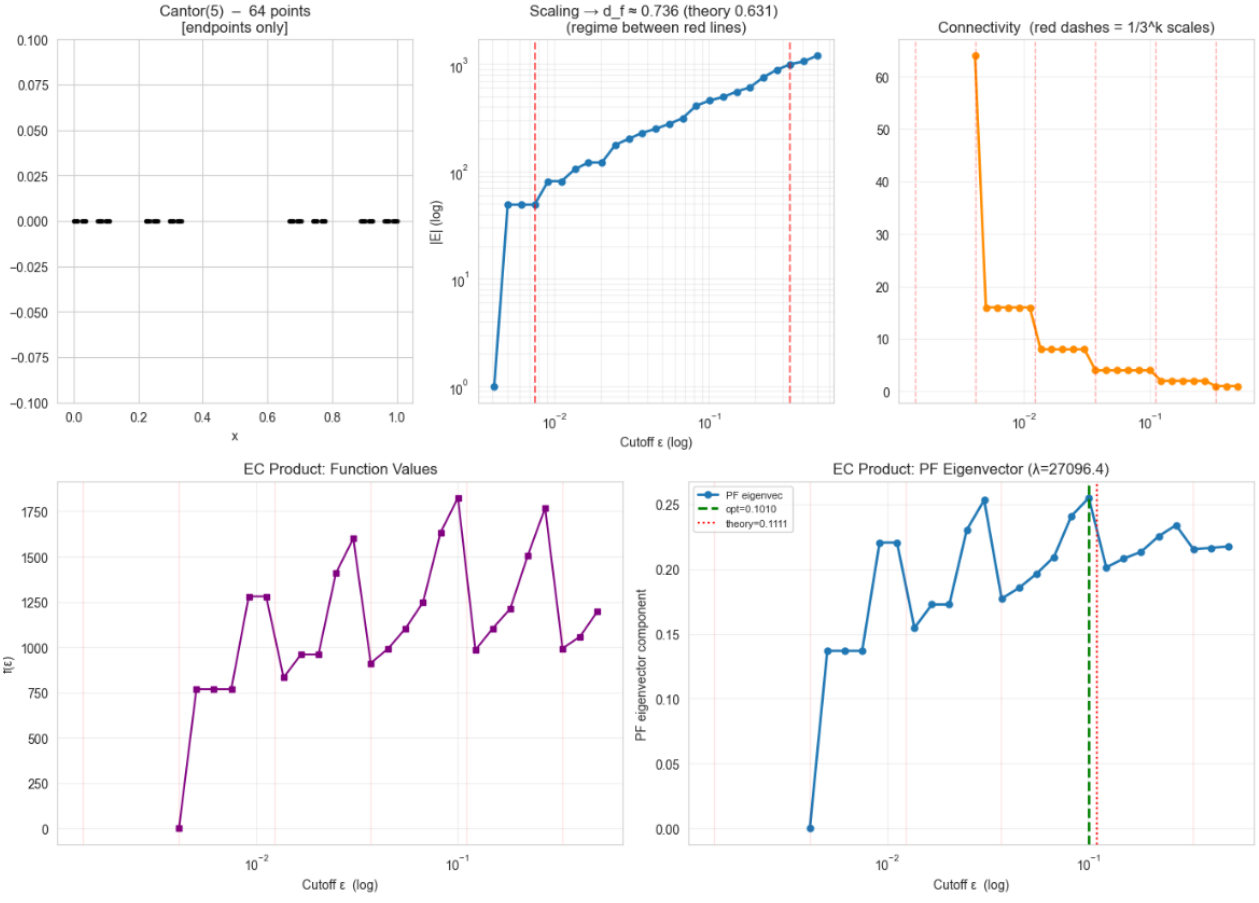
Generated 64 points on Cantor set
Number of intervals at level 5: 32

Theoretical Cantor scales: ['0.333333', '0.111111', '0.037037', '0.012346', '0.004115', '0.001372']

Cutoff range (log-spaced): [0.004115, 0.500000]

----------------------------------------------------------------------
Function                              λ_max   Opt cutoff  Closest scale   Ratio
----------------------------------------------------------------------
  EC Product                      27096.37     0.100951       0.111111   0.909

----------------------------------------------------------------------
Fractal Dimension Analysis:
----------------------------------------------------------------------
  Theoretical Cantor dimension : 0.6309
  Estimated from scaling (slope of linear regression, dim_lr_slope) : 0.7357
  Relative error  lr_slope_dim  : 16.6%
  Perron-Frobenius Cantor dimension pf_dim : 2.6699
  (Number of edges of optimal graph : 456, optimal cutoff : 0.1010)
  Relative error  pf_dim  : 323.2%
```



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======================================================================
CANTOR SCALING ANALYSIS  (n=5,  3 interior pts/interval)
======================================================================

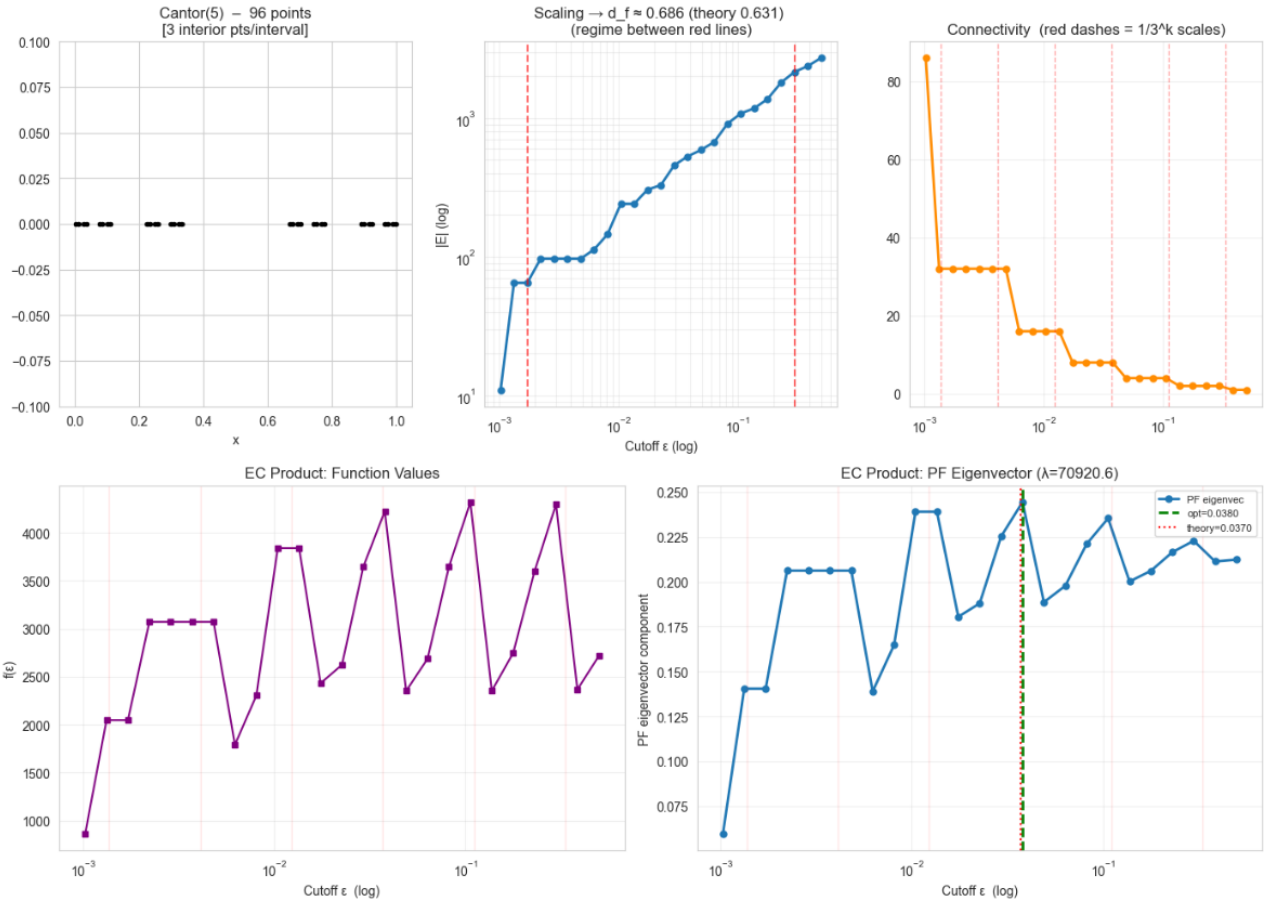
Generated 96 points on Cantor set
Number of intervals at level 5: 32

Theoretical Cantor scales: ['0.333333', '0.111111', '0.037037', '0.012346', '0.004115', '0.001372']

Cutoff range (log-spaced): [0.001029, 0.500000]

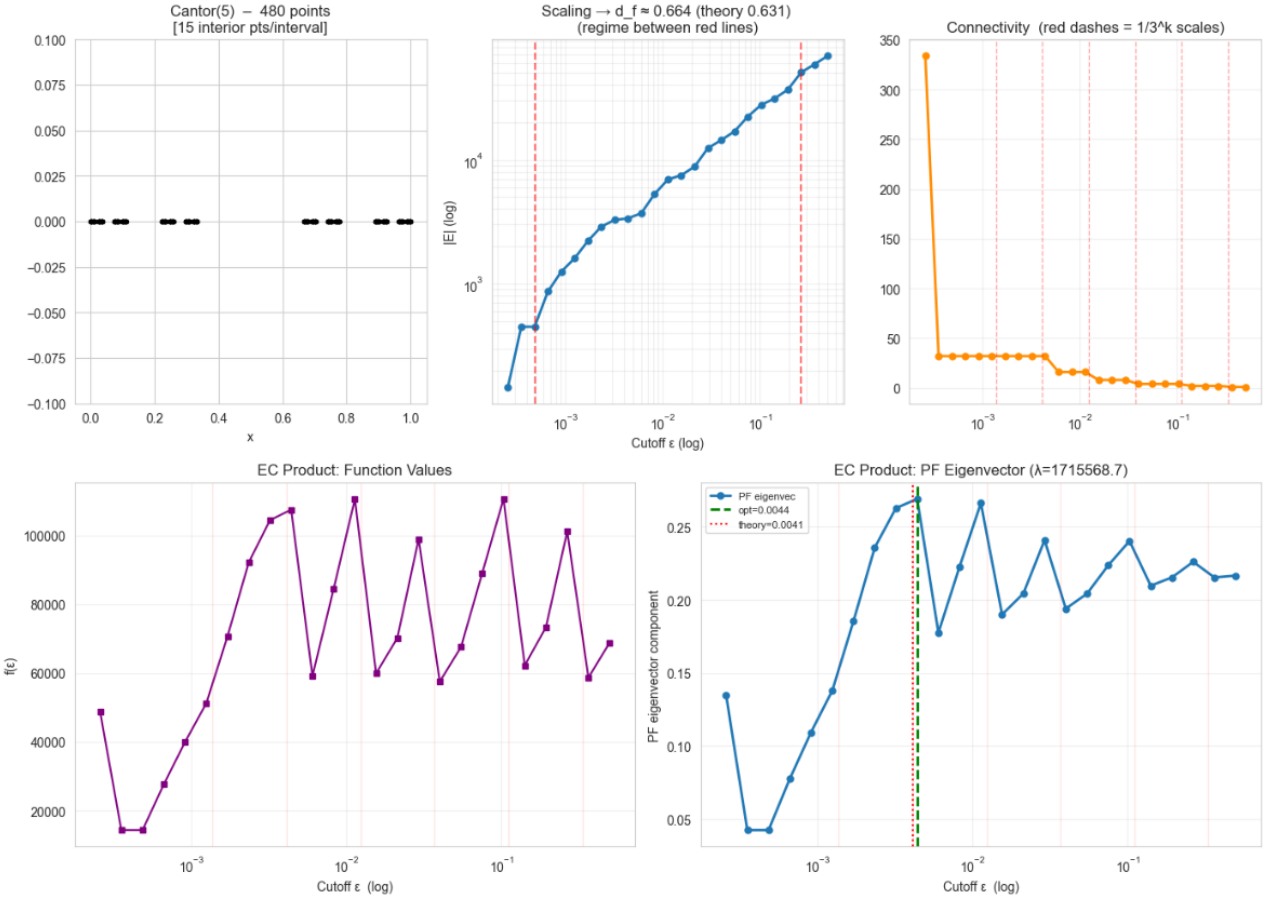
----------------------------------------------------------------------
Function                              λ_max   Opt cutoff  Closest scale   Ratio
----------------------------------------------------------------------
  EC Product                      70920.61     0.037979       0.037037   1.025

----------------------------------------------------------------------
Fractal Dimension Analysis:
----------------------------------------------------------------------
  Theoretical Cantor dimension : 0.6309
  Estimated from scaling (slope of linear regression, dim_lr_slope) : 0.6861
  Relative error  lr_slope_dim  : 8.8%
  Perron-Frobenius Cantor dimension pf_dim : 1.9167
  (Number of edges of optimal graph : 528, optimal cutoff : 0.0380)
  Relative error  pf_dim  : 203.8%
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======================================================================
CANTOR SCALING ANALYSIS  (n=5,  15 interior pts/interval)
======================================================================
Generated 480 points on Cantor set
Number of intervals at level 5: 32
Theoretical Cantor scales: ['0.333333', '0.111111', '0.037037', '0.012346', '0.004115', '0.001372']
Cutoff range (log-spaced): [0.000257, 0.500000]
----------------------------------------------------------------------
Function                            λ_max   Opt cutoff  Closest scale   Ratio
----------------------------------------------------------------------
  EC Product                   1715568.73     0.004401       0.004115   1.069

Fractal Dimension Analysis:
----------------------------------------------------------------------
  Theoretical Cantor dimension : 0.6309
  Estimated from scaling (slope of linear regression, dim_lr_slope) : 0.6640
  Relative error  lr_slope_dim  : 5.2%
  Perron-Frobenius Cantor dimension pf_dim : 1.4965
  (Number of edges of optimal graph : 3360, optimal cutoff : 0.0044)
  Relative error  pf_dim  : 137.2%
```




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======================================================================
CANTOR SCALING ANALYSIS  (n=7,  15 interior pts/interval)
======================================================================
Generated 1920 points on Cantor set
Number of intervals at level 7: 128
Theoretical Cantor scales: ['0.333333', '0.111111', '0.037037', '0.012346', '0.004115', '0.001372', '0.0004
57', '0.000152']
Cutoff range (log-spaced): [0.000029, 0.500000]
----------------------------------------------------------------------
Function                            λ_max   Opt cutoff  Closest scale   Ratio
----------------------------------------------------------------------
  EC Product                   33760675.43     0.000423       0.000457   0.925

Fractal Dimension Analysis:
----------------------------------------------------------------------
  Theoretical Cantor dimension : 0.6309
  Estimated from scaling (slope of linear regression, dim_lr_slope) : 0.6357
  Relative error  lr_slope_dim  : 0.8%
  Perron-Frobenius Cantor dimension pf_dim : 1.2238
  (Number of edges of optimal graph : 13440, optimal cutoff : 0.0004)
  Relative error  pf_dim  : 94.0%
```